

Mouth Guards : “An Orofacial Helmet”

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ABSTRACT

This review presents the literature dealing with oral-facial injuries received during participation in sport and the possibilities open to athletes for their prevention. In particular, the article examines five different aspects of this topic: the risk of dental injury while playing; the role of the mouthguard in preventing injury; types of athletic mouthguard; implications for patients undergoing orthodontic treatment and behavioural aspects of mouthguard wear.

Participation in a number of sports does carry a considerable risk of sustaining dental injury; not only in the so-called contact sports such as Rugby, Football, Boxing and Hockey, but also in less obviously dangerous sports such as Basketball. Although, some evidence exists to the contrary, the majority of studies have found the mouthguard to be the most effective way of preventing such injuries. While much progress has been made in this area, the profession could do much more to promote the greater use of mouthguards.

INTRODUCTION

This article reviews the literature relating to injuries sustained during participation in sporting activities and use of a mouthguard in preventing such injuries.

The mouthguard (also referred to as a *gumshield* or mouth protector) is defined as a *resilient device or appliance placed inside the mouth to reduce oral injuries, particularly to*

teeth and surrounding structures. Mouthguards are effective because they considerably diminish the deflection of teeth subjected to stress in comparison with the unprotected teeth. (Hoffman, After, Rudolph & Goz, 1999). Mouthguards are used to reduce injuries to the orofacial complex of participants in contact sports. Broken teeth, soft-tissue injuries, bone fractures and brain concussion are all reduced in wearers of

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mouthguards participating in sports which produce impacts between players and/or sporting implements (Hoffmann et al, 1999; Gellier, 1966; Wood, 1972 & Wei, 1974).

This review is divided into the following five sections (Newsome, Tran & Cook, 2001).

1. The risk of dental injury while playing sports,
2. The role of the mouthguard in preventing injury,
3. Types of athletic mouthguards,
4. Implications for patients undergoing orthodontic treatment and
5. Behavioural aspects of mouthguard wear.

THE RISK OF DENTAL INJURY WHILE PLAYING SPORT

It has been widely reported that participation in sports carries a considerable risk of sustaining dental injury. (Some & Ylipaavalnime, 1988). Contact sports are defined as those sports in which players physically interact with each other, trying to prevent the opposing team or person from winning.

The prevalence of injuries in certain contact sports, such as Rugby and Football, has received considerable attention in the dental literature. It has been estimated that players of such sports have a 10% chance, per playing season, of sustaining an injury and up to a 50% chance of injury in their playing career. Although, the incidence of dental trauma, compared with all accidents in contact team sports, is seldom very high, varying from 2% to 33% (Johnston & Messer, 1966; Heintz, 1968).

An almost universal finding is that the majority of injuries affect the upper jaw, with the maxillary incisors being most prone to injury, often accounting for as many as 80% of all cases. In addition to the damage caused by a traumatic impact to the dento-alveolar structures, damage can also result facial bone fracture and more seriously, neck or brain injury resulting from increased cranial pressure and deformation, with the most common cause of concussion in sport being a blow to the mandible.

In children, sports accidents reportedly account for 10-39% of all dental injuries and often involve teeth with incomplete root formation. Boys are more likely than girls to incur injuries; at a ratio of 1.5:3:1. The highest incidence occurs between the ages of 08 and 11 (Johnston & Messer, 1996).

THE ROLE OF THE MOUTHGUARD IN PREVENTING INJURY

The overwhelming majority of studies stress the significant protective values of mouthguards (De Wet, 1980). It was first introduced for boxers, primarily as a result of the pioneering work of a number of American dentists including Cohen & Borish and Cathcart.

Mouthguards are quite effective in the prevention of dento-alveolar injuries. Stenger and Chapman do suggest that, 'the use of mouthguards should be encouraged in all contact sports, as the most important value of the mouthguard is the concussion-saving effect following impact to the mandible. One should make the wearing of mouthguards compulsory in all contact

sports. Heintz reported that the mandatory use of mouth protectors had cut the injury rate to the face and mouth from 50% of all injuries to 0.5% (Heintz, 1968).

The term 'mouthguard' is universal and generic and includes a large range and variety of products, from over-the-counter models bought at sporting goods stores to professionally manufactured and dentist-prescribed custom-made mouthguards. It is clear that mouthguards, just like the athletes they are designed to protect, are not the same.

TYPES OF MOUTHGUARDS

Theoretically, a properly fitted mouthguard should be protective, comfortable, resilient, tear-resistant, odourless, tasteless, inexpensive, easy to fabricate and should not interfere with speech.

According to the Academy for Sports Dentistry's (ASD) definition, three different types of mouthguards are available now.

Type I

Stock mouthguards, bought over-the-counter and designed to be used without further modification to adjust to an individual's morphological characteristics. (Fig. 1)



Figure-1 : Type I- Stock mouthguard

Such mouthguards are no longer widely available.

Stock mouthguards are preformed rubber or vinyl devices. Because these mouthguards are preformed and worn as is, they must be held in place or in position by clenching the teeth together. Consequently, this type of mouthguard interferes with both speech and breathing, and therefore, is often deemed unacceptable for use in sports.

Type II

Mouth-formed boil-and-bite mouthguards.

Also available commercially, they are made from a thermoplastic material that is immersed in hot water and then formed in the mouth by the athlete using finger, tongue and biting pressure.

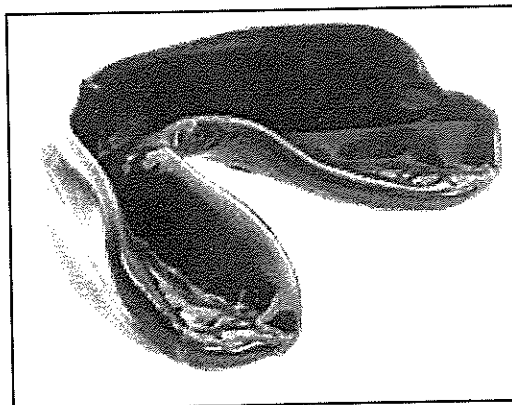


Figure-2 : Type II- Mouth-formed boil-and-bite mouthguard

While these devices offer better fit and retention, significant amount of thinning (between 70-90%) occurs at the occlusal surface (10) as a result of excessive or uncontrolled exertion of biting pressure.

Both Type 1 and 2 are commercially available and referred to as *over-the-counter* or on the shelf type. (Fig. 2)

Type III

Custom-made mouthguard, including the single layer and the multilayer laminated type. (Fig. 3)

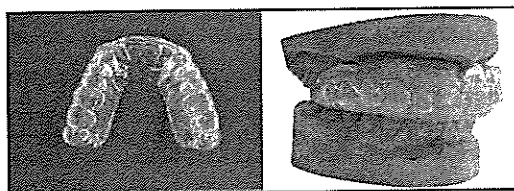


Figure-3 : Type III- Custom-made mouthguard

Several materials have been suggested for mouthguards, such as: (i) polyvinylacetate-polyethylene or ethylene vinyl acetate (EVA) copolymer, (ii) polyvinylchloride, (iii) silicon rubber, (iv) latex rubber, (v) acrylic resin and (vi) polyurethane. (Auroy, Duchatelard, Zmantar & Hennequin, 1996).

Bishop et al suggested that mouthguard materials should have low water sorption, high tear strength, good elasticity and compression behaviour, and high energy absorption characteristics. EVA copolymers met most of these requirements (Auroy et al, 1996).

The most common material is EVA Elastic properties are important for mouthguard material function. Elasticity of the copolymer determines the effectiveness of the mouthguard material, through the absorption of impact energy, as it is transmitted to underlying oral tissues.

The inclusion of air in the material enhances the energy absorption process, the sample with the largest air-inclusions showing the best energy absorption properties. The incorporation of air-cells in an EVA copolymer mouthguard material produced a reduction in transmitted forces when impacted by forces less than 10 kN. Reductions of transmitted forces of 32 per cent were observed. This material improvement could be utilized in stock, mouth-formed and vacuum-formed mouthguards (Westerman, storing fellow & Eccleston, 2002).

Custom-made mouthguards are by far the most expensive to fabricate, they do offer certain advantages that the others do not. These include optimal adaptation, maximum retention, superior comfort and minimal interference with both breathing and speech. Needless to say, professionally fitted mouthguards have a high acceptance rate.

Mouthguards have traditionally been made using vacuum moulding or pressure-formed and formed over a stone or plaster cast model of the dentition. Although there are certain notable differences between vacuum-formed and pressure-formed mouthguards, they are essentially constructed in a similar manner, that is, both rely on radiant heat transfer to allow for the thermoforming of the plastic material used to fabricate the mouthguards. Vacuum-formed mouthguards are generally fabricated using a single sheet of ethylene vinyl acetate (EVA) that is softened using low heat and then formed using low to moderate suction (vacuum) pressure. First,

the EVA sheet was softened for 3 min using the self-contained heat source. Upon expiration of the heating time, the assembly housing the heat source was swiveled away from its location over the EVA sheet, and the chamber enclosing the stone model and the softened mouthguard material was immediately pressurized to approximately $6 \cdot 10^5$ Pa (6 bar). As the chamber is pressurized, the mouthguard adapted to the shape of the stone model. The pressure was maintained for a total of 10 min. Once the EVA had cooled and set, the pressure was released and the mouthguard was trimmed and labeled (Del Rossi & Leyte-Vidal, 2007). Because of the limited heat and pressure that is used in the fabrication process, the shape of vacuum-formed mouthguards is typically short-lived. Once the shape of a mouthguard is lost, so too is fit, retention and comfort. An additional shortcoming of vacuum-formed mouthguards is related to the irregular distribution of mouthguard material, which can occur during the manufacturing process, resulting in a final product that is unevenly thick. It has also been demonstrated that when a vacuum machine is used, the EVA material stretches unevenly over the model and the incisal edges become dangerously thin. Protective thickness is important because as the *thickness of the mouthguard material increases logarithmically, the transmitted impact force decreases logarithmically*.

Alternatively, mouthguards can be pressure formed using high heat and high pressure. Because of the quality of adaptation

that results with this method of fabrication, pressure-formed mouthguards generally offer the best fit of all. Pressure-formed mouthguards can be laminated (i.e. multi-layered). Efficient and complete lamination cannot be achieved under low heat and vacuum. The layers will not properly fuse together with the vacuum machine, but will chemically fuse under high heat and pressure. The combination of the relatively high heat and pressure used in construction of the laminated mouthguard means that the mouthguard has virtually no elastic memory (Del Rossi & Leyte-Vidal, 2007).

IMPLICATIONS FOR PATIENTS UNDERGOING ORTHODONTIC TREATMENT

Only limited attention has been given to mouthguard use in those patients undergoing active orthodontic treatment. Most orthodontic treatment is carried out either just before or during the adolescent years, a period that as has already been shown, coincides with a peak in the incidence of dental trauma during participation in contact sports. Athletes undergoing orthodontic treatment may be at even higher risk as it has been shown that overjet and short upper lip are the main predisposing factors for dental trauma. It is also likely that oral-facial injuries will occur more easily when a child is wearing orthodontic appliances that may, for example, cause additional damage to the soft tissues following an accidental blow to the face. Aside from injuries to the patient, a blow to the appliance may result in loosening of the brackets and/or bending of the wires.

The factors thought to influence athletes' acceptance of mouthguards are comfort, wearability, stability, retention, the player's ability to speak and breathe etc. All of these factors are likely to be compromised further by the presence of orthodontic appliances, changing arch shape and the need for teeth to have unhindered movement during therapy. In orthodontic treatment, the teeth were moving at such a rate that the custom mouthguard do not fit properly for the entire 2 weeks. The stock mouthguard in these patients caused "vestibular irritation" (Newsome, Tran. & Cooke, 2001).

BEHAVIOURAL ASPECTS OF MOUTHGUARD WEAR

A commonly asked question by fabricators of mouthguards is "How thick should I make a mouthguard?" This question is more than likely a response to the realization that energy absorption and the resulting transmitted forces, in mouthguards, are linked to the thickness of the material and that wearer acceptance also often dictates the choice of a mouthguard.

The performance of mouthguards, in terms of energy absorption and transmitted forces, has been shown to improve with thickness or the inclusion of air-cells. Greater thickness detracts from wearer comfort while air-inclusions increase energy absorption without increasing thickness. Thicker mouthguards reduce the capacity for speech as well as interfere with respiratory efficiency (Westerman, String fellow & Eccleston, 2002).

Single-sheet EVA materials with thicknesses between 3 mm and 5 mm are commonly used in the fabrication of custom-made mouthguards. Thickness can be adjusted by removing materials during finishing or limiting thinning during manufacture. The pressure - laminated manufacturing process allows the thickness of finished mouthguards to be adjusted with extra laminates or by material removal during formation (Westerman, String fellow & Eccleston, 2002).

Effect of Colour

Of the two types of custom-fabricated mouthguards, pressure-formed protectors are preferred. Vacuum-formed mouth guards are shaped using low to moderate heat, whereas pressure-formed mouthguards are fashioned using high heat. This difference in heat could mean the difference between achieving proper forming temperature or not. Another factor that may determine whether or not the critical temperature is reached is color. Given that different colors absorb radiant heat at different levels, this effect may influence the final temperature attained by the plastic material, which in turn, may impact the level of adaptation that is achieved during the fabrication process.

As light-colored materials are likely to reflect a significant amount of energy striking its surface, a much lesser build up of heat would be expected within the substance. Conversely, dark-colored materials would be expected to provide the best adaptation as they absorb the greatest amount of infrared energy (Del Rossi, Lisman & Signorile, 2008).

If a light colored or transparent material is used to fabricate mouthguards, the users of these products should be informed that the shape of the mouthguards might not be maintained for an extended period of time, and therefore, the characteristic fit and retention afforded by these devices might be short lived.

According to Del Rossi et al (2007), a mouthguard made from darker-colored material is likely to fit more firmly and remain retentive for a longer period of time. This may increase user compliance and decrease the chances of displacing the mouthguard with contact.

Factors Affecting The Thickness

The fabrication of mouthguards is often a compromise between thickness and wearer comfort. Thicker mouthguards are often met with wearer resistance because of discomfort from lip and cheek displacement, speech interference and also respiratory restrictions. At the same time, very thin mouthguards are well accepted by users but are far less efficient in terms of energy absorption and transmitted forces.

It is well-documented that the actual final thickness of a mouthguard is fairly important in reducing the transmission of impact forces to the teeth (Newsome, Tran, & Cooke, 2001; Del Rossi & Leyte-Vidal, 2007). Impact testing has repeatedly demonstrated an inverse relationship between material thickness and force transmission. Park et al reported that by decreasing the thickness of a mouthguard by just half a millimeter (i.e. from 1.5 to 1.0 mm), the peak impact forces generated increased by roughly

48% (Del Rossi & Leyte-Vidal, 2007).

Clearly, every attempt must be made to keep the amount of thinning and the concomitant reduction in protection that results to a minimum.

Efficacy of mouthguards

Cohenca et al (2007) examined reported injuries to determine the effectiveness of mouthguards in reducing the incidence or severity of dental injuries, and found the incidence rate of injury was five times higher in players not using mouthguard (Cohenca, Roges & Roges, 2007).

Occlusion

Since mouthguards cover the occlusal portion of dental arch, the occlusion of mouthguards is another important factor for improving their efficacy.

Takeda et al (2008) discussed the influence of anterior occlusion of mouthguards in protecting the maxillary anterior teeth, indicating that occlusal contacts of the anterior part of mouthguard are necessary for reducing the impact force and tooth distortion (Tabeda et al, 2008).

Promotion of mouthguards

In order to increase the usage of mouthguards among athletes, it is of the utmost importance to increase the recognition and understanding of safety measures among not only players but also coaches and related associations. In those sports activities with frequent contacts such as Rugby, Basketball and so on, mandating of mouthguard usage, in certain age groups, should be seriously considered.

CONCLUSION

It is clear that the literature views mouthguards as offering considerable protection against sports related dental injuries. Equally it is clear that although custom-made mouthguards are seen to offer better protection than the 'boil-and-bite' type. Many people do not use them because they are perceived to be expensive and necessitate a visit to the dentist. In order to achieve the precise fit of the mouthguard, factors

related to the thermoforming process, such as working model residual moisture, heating temperature, color and so on, should be controlled. There is growing opinion that while much progress has been made in the use of mouthguards, the profession could do much more to promote their use, for example by targeting not only players but also governing bodies, coaches and schools; and also by finding ways to make mouthguards more affordable to the public.

REFERENCES

- Auroy, P., Duchatelard, P., Zmantar, N. & Hennequin, M. (1996). Hardness and shock absorption of silicone rubber for mouth guards. *Journal of Prosthetic Dentistry* 75 : 463-471.
- Cohenca, N., Roges, R.A. & Roges, R. (2007). The incidence and severity of dental trauma in inter collegiate athletes. *J Am Dent Assoc* 138 : 1121-6.
- De Wet, F. (1980). Prevention of Orofacial Sports Injuries in the Adolescent, Department of Conservative Dentistry and Paedodontics. Faculty of Dentistry, University of Pretoria, South Africa, 31: 313-317.
- Del Rossi, G. & Leyte-Vidal, M.A. (2007). Fabricating a better mouthguard. Part I : Factors influencing mouthguard thinning. *Dental Traumatology* 23: 149-154.
- Gelbier, S. (1966). The use and construction of mouth and tooth protectors for contact sports. *BrDentJ* 120:533-7.
- Gianluca Del Rossi, Peter Lisman & Joseph Signorile (2008). Fabricating a better mouthguard Part II. The effect of color on adaptation and fit. *Dental Traumatology* 24: 197-200.
- Heintz, W. (1968). Mouth protectors: a progress report. *Journal of the American Dental Association*; 77: 632-636.
- Hoffmann, J., Alfter, G., Rudolph, N. & Goz G. (1999). Experimental comparative study of various mouthguards. *Endodontics and Dental Traumatology*; 15: 157-163.
- Johnston, T. & Messer, L.B. (1996). An *in vitro* study of the efficacy of mouthguard protection for dentoalveolar injuries in deciduous and mixed dentitions. *Endod Dent Traumatol*; 12: 277-285.

- P. R. H. Newsome, D. C. Tran & M. S. Cooke (2001).** The role of the mouthguard in the prevention of sports-related dental injuries: a review. *International Journal of Paediatric Dentistry*; 11: 396-404 .
- Sane, J. & Ylipaavalniemi, P. (1988).** Dental trauma in contact team sports. *Endodontics and Dental Traumatology*; 4: 164-169.
- Takeda, T., Ishigami, K., Nakajima, K., Naitoh, K., Kurokawa, K., Handa, J. et al (2008).** Are all mouthguards the same and safe to use? Part II. The influence of anterior occlusion against a direct impact on maxillary incisors. *Dent Traumatol* 24:360-5.
- Wood, A.W. (1972).** Head protection - cranial, facial and dental in contact sports. *Oral Health* 62:23-33.
- Wei, S.H. (1974).** Prevention of injuries to anterior teeth. *Int Dent J.* 24:30-49.
- Westerman, B., Stringfellow, P.M. & Eccleston, J.A. (2002).** EVA mouthguards: how thick should they be? *Dent Traumatol* 18: 24-27.

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